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Reliability evaluation for a manufacturing network with multiple production lines ☆

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ABSTRACT

This paper focuses on performance evaluation of a manufacturing system with multiple production lines based on the network-analysis perspective. Due to failure, partial failure, or maintenance, the capacity of each machine is stochastic (i.e., multi-state). Hence, the manufacturing system can be constructed as a stochastic-flow network, named manufacturing network herein. This paper intends to measure the probability that the manufacturing network can satisfy customers' orders. Such a probability is referred to as the system reliability. A graphical representation is first proposed to transform a manufacturing system into a manufacturing network. Thereafter, we decompose the manufacturing network into general processing paths and reworking paths. Three algorithms are subsequently developed for different scenarios and multiple production lines to generate the minimal capacity vectors that machines should provide to satisfy demand. The system reliability can be derived in terms of such capacity vectors afterwards.

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1. Introduction

In a manufacturing system, each production line consists of a sequence of machines and each machine has a specific function. Multiple production lines can be utilized to produce products with higher efficiency and flexibility (Chen & Lan, 2001; Lan, 2007). From the perspective of production management and decision making, it is a critical issue to construct a manufacturing system with multiple production lines and to evaluate its performance. Network analysis is an achievable approach to be applied for evaluating the performance of a manufacturing system. That is, a manufacturing system can be modeled as a manufacturing network and thus further network analysis can be implemented. A network-based approach proposed by Chen and Lan (2001) has been studied to design a manufacturing system producing the same product type in multiple production lines. Their work discussed a particular case in which products are sold at once when they are produced. Thereafter, Lan (2007) proposed a network model to construct a manufacturing system with multiple production lines to maximize profit.

Several researchers have also studied performance evaluation for a manufacturing system by adopting network analysis (Lee & Garcia-Diaz, 1993, 1996; Listeş, 2007; Paquet, Martel, & Montreuil, 2008). Based on flow network concept, Lee and Garcia-Diaz (1993, 1996) proposed an approach to solve the grouping problem for a manufacturing system by measuring the functional similarity between machines as a performance indicator. Listeş (2007) designed a generic stochastic model for supply-and-return network in a closed-loop system to locate the general manufacturing path and reworking path. Paquet et al. (2008) proposed an optimization methodology to design the manufacturing network producing several products, in which the deterministic capacities of processors and workers are considered. Previous works, however, did not emphasize the stochastic capacity of each machine for evaluating the capability of a manufacturing network. Those works mainly focused on costs, profits, and sales. The capability of a manufacturing network reflects the order satisfaction ability directly. Nevertheless, evaluations of costs, profits, and sales mainly provide the financial information rather than the production performance.

The failure rate of each machine influences the capability of a manufacturing network and a machine failure leads to defective products, which might be reworked or scrapped. (Liu, Kim, & Hwang, 2009; Pillai & Chandrasekharan, 2008). Thus, an important issue to be considered is how the reworking action affects the amount of output products in a manufacturing system. In many cases, defective products still have substantial value, e.g. caused by expensive input materials, and thus there is an incentive to rework those products to the 'as new' condition (Buscher & Lindner, 2007). In several applications, the reworking action is implemented with the same machines for producing products. This implies that a manufacturing network would have two sources, from the general manufacturing path(s) and the reworking path(s), for satisfying demand (Buscher & Lindner, 2007; Teunter, Kaparis, & Tang, 2008).

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In a manufacturing network, each edge can be regarded as a machine and each node is an inspection station after the machine. Due to failure, partial failure, or maintenance, each machine in the manufacturing network has stochastic capacity levels (i.e., multi-state) with performance ranging from a perfectly working level to complete malfunction. Therefore, the manufacturing network is also multi-state and we can treat it as the so-called stochastic-flow network (Alexopoulos, 1995; Aven, 1985; Hudson & Kapur, 1985; Xue, 1985; Lin, 2004; Lin, 2009a, 2009b, 2009c; Lin & Chang, 2011a, 2011b; Zuo, Tian, & Huang, 2007). To assess the capability that a manufacturing network satisfies the customers' requirements, Lin (2009a) focused on two-commodity reliability evaluation of a manufacturing network in terms of minimal path (MP), where an MP is a path whose proper subsets are no longer paths. That is, an MP is a sequence of edges from source to sink which contains no cycle, and which carries sufficient capacity (Zuo et al., 2007). In Lin's work, the system reliability is defined as the probability that the manufacturing network satisfies two-commodity demand. A great deal of research (Lin, 2004; Lin, 2009a; Lin, 2009b, 2009c; Lin & Chang, 2011a, 2011b; Zuo et al., 2007) has also been devoted to studying the system reliability of a stochastic-flow network (such as manufacturing, computer, and communication networks) in terms of MP. In these studies, the demand transmitted through a network must obey flow conservation (Ford & Fulkerson, 1962), which implies that no flow will be increased or decreased during transmission. More importantly, some characteristics related to the manufacturing network such as reworking and scraps were not considered in these papers. For a practical manufacturing network, the input flow processed by each machine would not be the same as output flow due to the non-zero failure rate of each machine. That is, the output products of the manufacturing network might be less than the input raw materials. It implies that the traditional methodology for the stochastic-flow network problem could not be applied in such a case due to the violation of flow conservation. Moreover, based on the MP concept, an edge (machine) would not appear on the same path more than one time; otherwise it would not be an MP. However, defective WIP (work-in-process) from a machine would be reworked starting from a previous machine(s) or the same machine(s) (Buscher & Lindner, 2007; Teunter et al., 2008), which violates the basic concept of MP.

The existing stochastic-flow network models developed for reliability evaluation and mentioned above cannot deal with unbalanced flow and non-MP path. However, a manufacturing system is not a typical network from the aspects of flow (due to failure) and path (due to rework). To overcome these limitations, we propose a graphical-based methodology to transform the manufacturing system into a manufacturing network firstly. The transformed network topology is decomposed into several paths, the general manufacturing path(s) and the reworking path(s). This graphical representation is beneficial for determining the sources (from the general manufacturing path or the reworking path) and amount of input flow for each machine. Based on these decomposed paths, we derive the minimal capacity vectors that machines should provide to fulfill demand, in which the demand is d units of product produced per unit time. In terms of such minimal capacity vectors, we evaluate the probability that the manufacturing network could produce d by applying stochastic-flow network model. Such a probability is referred to as system reliability, which is a performance indicator to measure the possibility that the manufacturing network satisfies customers' orders. The remainder of this paper is organized as follows. Problem description for two scenarios and assumptions are proposed in Section 2. Scenario 1 considers identical production lines in parallel producing the same product type; while Scenario 2 considers two distinct production lines producing the same product type. A graphical-based methodology to transform and to decompose the manufacturing system is described in Section 3. Model construction and system reliability evaluation are formulated in Section 4. Two algorithms are proposed in Section 5 to generate the minimal capacity vectors for demand satisfaction in both scenarios. Numerical examples in the context of a tile manufacturing system and discussion on results are demonstrated in Section 6. A generic model for the manufacturing system with multiple production lines is proposed in Section 7. Conclusion is summarized in Section 8.

2. Problem description and assumptions

2.1. Problem description

A flow-shop manufacturing system in which products are make-to-stock (MTS) is considered in this paper. That is, this manufacturing system is a high-volume system with standardized machines and processes to produce identical or highly similar products (Framinan, 2007; Stevenson, 2007). Products produced by such a manufacturing system are then provided to customers from finished products stock (Stevenson, 2007). To analyze the manufacturing network, we emphasize the input flow of each machine, in which the input flow is defined as the input amount that each machine processes per unit time. Under the situation that WIP may be defective during processing, such a manufacturing system is requested to produce d units of product per unit time. In the manufacturing system, products are produced by multiple production lines. For convenience, we concentrate on two production lines case firstly. The proposed methodologies and algorithms can then be easily extended to more than two production lines case in Section 7. Two scenarios are discussed in this paper, say.

Scenario 1: Two identical production lines in parallel producing the same product type. That is, the ordered machines and their functions in both two production lines are the same.

Scenario 2: Two distinct production lines producing the same product type. In this case, most machines and their functions in both two production lines are the same. However, one of the two lines may produce a compact product type by fewer processes (machines). Another line produces more than one product types (regular product type by all machines and compact product type by fewer machines). Thus, a common machine(s) is used when producing regular product type in both lines.

2.2. Assumptions

To evaluate the system reliability of a manufacturing system, some assumptions are addressed as follows:

1. Each node (inspection station) is perfectly reliable. It indicates that the inspection station would not damage any WIP/products.
2. The capacity of each edge (machine) is a random variable according to a given probability distribution.
3. The capacities of different edges (machines) are statistically independent.
4. No assembling action is taken for producing the product and one material can produce one product.
5. Each defective WIP is reworked at most one time by the same machine. It implies that such a defective WIP is repaired until a usable state. If the defective WIP after reworking is still defective, such a defective WIP is non-reparable and is scrapped.

3. Methodology

In this section, we propose a common methodology for both scenarios to construct a manufacturing system by a topology transformation. A graphical-based technology is utilized to transform the manufacturing system into a manufacturing network. Based on this manufacturing network, we decompose it into several paths that are benefit to further network analysis.

3.1. Graphical-based transformation

We utilize an AOA (activity-on-arrow) network diagram approach to transform a manufacturing system into a manufacturing network. Each edge (arrow) is regarded as a machine and each node denotes an inspection station following the machine. Let (**N**, **E**) be a manufacturing network with two production lines, where **N** represents the set of nodes and $\mathbf{E} = \{e_i | i = 1, 2, \ldots, n_1 + n_2\}$ represents the set of edges; n_1 is the number of machines in the first production line P_1 and n_2 is the number of machines in the second line P_2. For instance, Fig. 1 represents a manufacturing system with two production lines, say P_1 (consisting of e_1, e_2,e_3, and e_4) and P_2 (consisting of e_5, e_6, e_7, and e_8), producing the same product type. Each production line consists of four machines ($n_1 = n_2 = 4$) and the success rate of each machine is p, where $0 \leqslant p \leqslant 1$ (i.e., the failure rate is $q = 1 - p$). The reworking action is that defective WIP output from e_3 and e_7 can be reworked staring from e_2 and e_6, respectively. In Fig. 1, the two black nodes in each production line represent the input inspection (check for raw materials) and output inspection (check for final products), respectively. White nodes between two machines denote the inspection for WIP to check if it can enter the next process (success) or should be scrapped (failure). The meshed nodes means that defective WIP (output from e_3 or e_7) inspected by this inspection station can be reworked. Two notations are involved in Fig. 1, in which I_j is the input amount of raw materials and O_j is units of output product for jth production line P_j, respectively (j = 1, 2).

However, we can not distinguish the input flows from the regular process (without reworking) or the reworking process from Fig. 1. Thus, we transform the manufacturing system into a manufacturing network as Fig. 2, in which a dummy-machine e'_i is set to denote the machine e_i doing the reworking action. The transformed network is helpful for further analysis because it is easily to distinguish the input amount from regular process and reworking process of each machine. Fig. 2 shows the input flow under each edge. Thus, we obtain the input flows (from the regular process and the reworking process) for each machine clearly.

3.2. Decomposition method

To analyze the manufacturing network in terms of paths, we decompose each production line into two paths, named the general manufacturing path $P_j^{(G)}$ and the reworking path $P_j^{(R|r,r-k)}$, where r and k are the indices denoting that defective WIP output from the rth sequenced machine are reworked starting from previous k machine(s). Take the same example in Fig. 2 for instance, the set $\{e_1, e_2, e_3, e_4\}$ in production line P_1 would be a general manufacturing path $P_1^{(G)}$. On the other side, the other path with reworking action is $P_1^{(R|3,3-1)} = \{e_1, e_2, e_3, e'_2, e'_3, e'_4\}$, where $(R|3,3\text{-}1)$ denotes that defective WIP output from the third sequenced ($r = 3$) machine e_3 in the production line P_1 is reworked starting from previous one ($k = 1$) machine (i.e., starting from e_2). For the special case that k is set to be zero, it implies that the defective WIP is reworked at the same machine (rth sequenced machine). In fact, no defective WIP is processed by e_1, e_2, and e_3 and the input flow would be zero for these edges. Thus, e_1, e_2, and e_3 are ignored and merely the machines e'_2, e'_3, and e'_4 doing reworking action would be retained. Since the machine e_i and the dummy-machine e'_i are the same, the reworking path $P_1^{(R|3,2)}$ is simplified to be $\{e_2, e_3, e_4\}$ such as it is shown in Fig. 3. The dashed lines indicate that those machines do not process the reworking action. Similarly, we apply the same procedure to decompose P_2 into $P_2^{(G)} = \{e_5, e_6, e_7, e_8\}$ and $P_2^{(R|3,3-1)} = \{e_6, e_7, e_8\}$.

4. Model construction

In this section, we construct a common model for both scenarios to derive the input amount of raw materials and input flow of each machine in the manufacturing network. Hence, the loading of each machine and the minimal capacity each machine should provide can be determined according to the input amount derived. Thereafter, the system reliability of the manufacturing network is evaluated in terms of minimal capacity vectors.

4.1. Determination of input amount

In order to produce sufficient products for satisfying demand d, the input amount of raw materials should be pre-determined. For each production line P_j, suppose that I_j units of raw materials are expected to produce O_j units of product; we intend to obtain the relationship between I_j and O_j such that $\sum_{j=1}^{2} O_j$ satisfies $\sum_{j=1}^{2} d_j = d$, where d_j is the assigned demand for production line P_j.

Let n_j denote the number of machines in production line P_j. We thus have the amount of output products $O_j = (I_j p^{n_j} + I_j p^{n_j+k} q)$, where the first term $I_j p^{n_j}$ is produced by the general manufacturing path $P_j^{(G)}$ while the second term $I_j p^{n_j+k} q$ is processed through the reworking path $P_j^{(R|r,r-k)}$. The following equation guarantees the manufacturing network can produce exact sufficient output O_j that satisfies demand d_j.

$$I_j = d_j / (p^{n_j} + p^{n_j+k} q), \quad j = 1, 2. \tag{1}$$

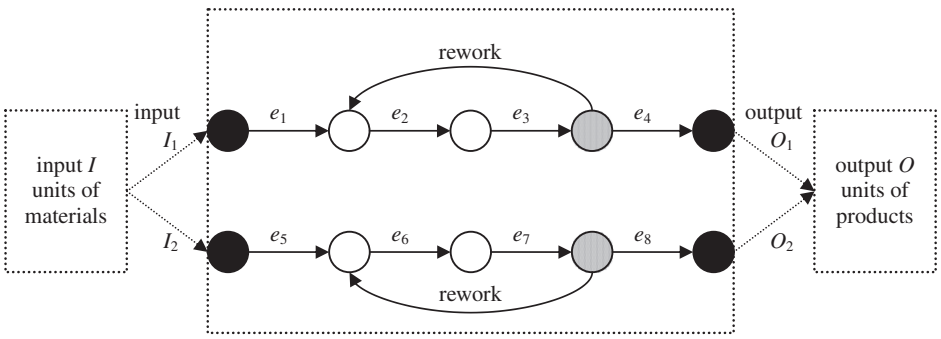


Fig. 1. Two identical production lines in parallel for Scenario 1.

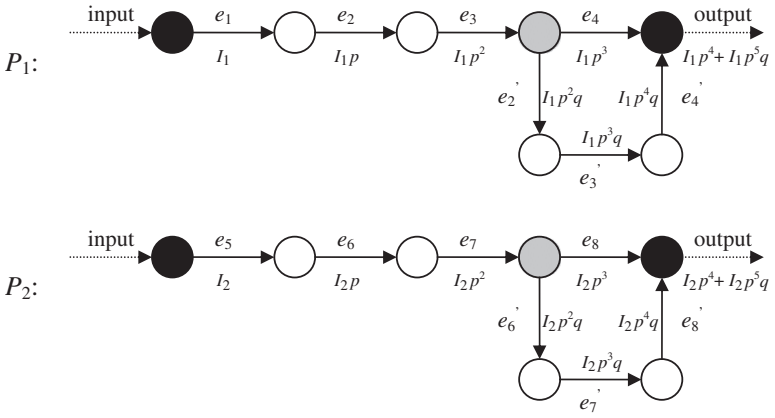


Fig. 2. The transformed manufacturing network for Fig. 1.

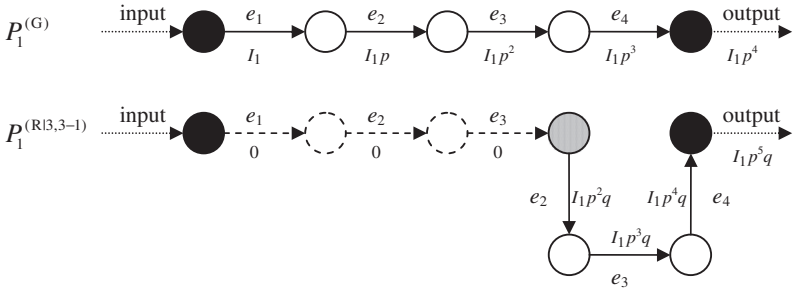


Fig. 3. Decomposition for P_1.

4.2. Determination of input flow for each machine

To determine the input flow of each machine, we utilize $f_{j,i}^{(G)}$ and $f_{j,i}^{(R|r,r-k)}$ to represent the input amount of the ith machine in the jth production line for $P_j^{(G)}$ and $P_j^{(R|r,r-k)}$, respectively. The input raw materials/WIP processed by the ith machine e_i should satisfy the following constraint,

$$\sum_{j=1}^{2}\left(f_{j,i}^{(G)}+f_{j,i}^{(R|r,r-k)}\right)\leqslant M_i,\quad i=1,2,\ldots,n_1+n_2. \tag{2}$$

Constraint (2) ensures that the total amount of input flow entering machine e_i does not exceed the maximal capacity M_i. The term $\sum_{j=1}^{2}\left(f_{j,i}^{(G)}+f_{j,i}^{(R|r,r-k)}\right)$ is further defined as the loading of each machine, say l_i, and we have the following equation,

$$l_i=\sum_{j=1}^{2}\left(f_{j,i}^{(G)}+f_{j,i}^{(R|r,r-k)}\right),\quad i=1,2,\ldots,n_1+n_2. \tag{3}$$

Let x_i denote the capacity of each machine e_i. According to assumption 2, the capacity x_i of each machine e_i is a random variable and thus the manufacturing network is stochastic, where x_i takes possible values $0=x_{i1}<x_{i2}<\ldots<x_{ic_i}=M_i$ for $i=1, 2, \ldots, n_1+n_2$ with c_i denoting number of possible capacities of e_i. Under the state $X=(x_1,x_2,\ldots,x_{n_1+n_2})$, constraint (4) is necessary to guarantee that e_i can process the input raw materials/WIP

$$x_i\geqslant l_i=\sum_{j=1}^{2}\left(f_{j,i}^{(G)}+f_{j,i}^{(R|r,r-k)}\right)\quad \text{for } i=1,\ 2,\ \ldots,\ n_1+n_2. \tag{4}$$

4.3. Evaluation of system reliability

Given the demand d, the system reliability R_d is the probability that the output product from the manufacturing network is not less than d. Thus, the system reliability is $\Pr\{X|V(X)\geqslant d\}$, where $V(X)$ is defined as the maximum output under X. It implies that each machine should provide enough capacity to process the input raw materials/WIP and finally produce enough units of output products. However, it is not a wise way to find all X such that $V(X)\geqslant d$ and then cumulate their probabilities to derive R_d. Any minimal vector Y in the set $\{X|V(X)\geqslant d\}$ is claimed to be the minimal capacity vectors for d. That is, Y is a minimal capacity vector for d if and only if (i) $V(Y)\geqslant d$ and (ii) $V(Y')<d$ for any capacity vectors Y' such that $Y'<Y$. Given $Y_1, Y_2, \ldots, Y_h$, the set of minimal capacity vectors satisfying demand d, the system reliability R_d is

$$R_d=\Pr\left\{\bigcup_{v=1}^{h}B_v\right\}, \tag{5}$$

where $B_v=\{X|X\geqslant Y_v\}$, $v=1, 2, \ldots, h$. Several methods such as the Recursive Sum of Disjoint Products (RSDP) algorithm (Zuo et al., 2007; Lin & Chang, 2011a, 2011b), inclusion–exclusion method (Hudson & Kapur, 1985; Lin, 2004, 2009a, 2009b, 2009c; Xue, 1985), disjoint-event method (Hudson & Kapur, 1985; Yarlagadda & Hershey, 1991), and state-space decomposition (Alexopoulos, 1995; Aven, 1985) may be applied to compute $\Pr\{\bigcup_{v=1}^{h}B_v\}$. The inclusion–exclusion method quickly becomes computationally intractable, especially when the size of the network is large (Zuo et al., 2007). In practice, the RSDP algorithm demonstrates better computational efficiency than the state-space decomposition approach for large networks (Zuo et al., 2007). Hence, the RSDP algorithm is chosen as the method to derive system reliability herein.

5. Algorithms to generate the minimal capacity vectors for *d*

5.1. Algorithm I: For Scenario 1

In Scenario 1, there are two production lines in parallel, $P_1=\{e_1,e_2,\ldots,e_{n_1}\}$ and $P_2=\{e_{n_1+1},e_{n_1+2},\ldots,e_{n_1+n_2}\}$, producing

the same product type ($n_1 = n_2$). Defective WIP output from the rth sequenced machine is reworked starting from the previous k machine (i.e., from the $(r-k)$ th sequenced machine). The minimal capacity vectors for d can be derived as following steps:

Step 1: Find the maximum output for each path.

$$O_{1,\max} = \min\{M_i(p^{t_{j,i}+1} + \alpha_i p^{t_{j,i}+k+1} q) | i \in P_1\} \text{ and}$$
$$O_{2,\max} = \min\{M_i(p^{t_{j,i}+1} + \alpha_i p^{t_{j,i}+k+1} q) | i \in P_2\}, \quad (6)$$

where $t_{j,i}$ is the number of follow-up machines after machine e_i on the general manufacturing path $P_j^{(G)}$. The indicator α_i equals 1 for the ith machine not on the reworking path; otherwise α_i equals 0.

Step 2: Find the demand assignment (d_1, d_2) satisfying $d_1 + d_2 = d$ under constraints $d_1 \leqslant O_{1,max}$ and $d_2 \leqslant O_{2,max}$.

Step 3: For each demand pair (d_1, d_2), do the following steps.

3.1. Determine the amount of input materials for each production line by

$$I_1 = d_1/(p^{n_1} + p^{n_1+k} q) \quad \text{and} \quad I_2 = d_2/(p^{n_2} + p^{n_2+k} q), \quad (7)$$

3.2. Determine the input flows for each machine e_i. For $e_i \in P_j$, the input flows are derived as follows,

$$f_{j,i}^{(G)} = I_j p^{s_{j,i}-1} \text{ for } i \text{ such that } e_i \in P_j^{(G)} \text{ and}$$
$$f_{j,i}^{(R|r,r-k)} = I_j p^{s_{j,i}+k-1} q \text{ for } i \text{ such that } e_i \in P_j^{(R|r,r-k)}, \quad (8)$$

where r is the sequence of the machine whose defective output can be reworked; $s_{j,i}$ is the sequence of e_i on the general manufacturing path $P_j^{(G)}$.

3.3. Transform input flows from general manufacturing paths and reworking paths into machines' loading vector $L = (l_1, l_2, \ldots, l_{n_1+n_2})$ via

$$l_i = \sum_{j=1}^{2} \left(f_{j,i}^{(G)} + f_{j,i}^{(R|r,r-k)} \right). \quad (9)$$

3.4. For each machine, find the smallest possible capacity x_{ic} such that $x_{ic} \geqslant l_i > x_{i(c-1)}$. Then $Y = (y_1, y_2, \ldots, y_{n_1+n_2})$ is a minimal capacity vector for d, where $y_i = x_{ic}$ for all i.

Step 4: Those Y obtained in Step 3 are the minimal capacity vectors for d.

The following theorem guarantees that all Y generated from the proposed algorithm are the minimal capacity vectors for d.

Theorem 1. *The capacity vector Y generated from the algorithm is a minimal capacity vector for d.*

Proof. Suppose that Y is not a minimal capacity vector for d, then there exists a minimal capacity vector Z for d such that $Z < Y$. Without loss of generality, we set $Z = (z_1, z_2, \ldots, z_i, \ldots, z_{n_1+n_2})$ and there exists at least one $z_i < y_i$. The situation $z_i < y_i$ implies that $z_i < \sum_{j=1}^{2} \left(f_{j,i}^{(G)} + f_{j,i}^{(R|r,r-k)} \right)$ and cannot afford enough capacity for the input units of WIP which contradicts the assumption that Z is the minimal capacity vector for d (note that $l_i = \sum_{j=1}^{2} \left(f_{j,i}^{(G)} + f_{j,i}^{(R|r,r-k)} \right)$ and y_i is the minimal capacity satisfying l_i). Thus, we conclude that Y generated from the algorithm is a minimal capacity vector for d. □

5.2. Algorithm II: For Scenario 2

In Scenario 2, there are two production lines with a common machine e_c installed in P_1, we thus have $P_1 = \{e_1, e_2, \ldots, e_c, \ldots, e_{n_1}\}$ and $P_2 = \{e_{n_1+1}, e_{n_1+2}, \ldots, e_c, \ldots, e_{n_1+n_2}\}$, for producing the same product type ($n_1 > n_2$). For instance as shown in Fig. 4, we have a path $P_1 = \{e_1, e_2, e_3, e_4\}$ which can produce both the regular (by all machines) and the compact (by machines e_1, e_3, e_4) product types. Another path $P_2 = \{e_5, e_7, e_8\}$ only produces the compact product type, where e_5 (resp. e_7 and e_8) operates the same function as e_1 (resp. e_3 and e_4). However, once the production manager decides to produce the regular product type by both production lines, it is essentially that e_2 serves in both production lines. In that case, the common machine $e_c = e_2$ and $P_2 = \{e_5, e_2, e_7, e_8\}$. Defective WIP output from the rth sequenced machine is reworked starting from the previous k machine (i.e., from the $(r-k)$th sequenced machine). Given the success rate of each machine is set as p, the minimal capacity vectors for d can be derived as following steps.

Step 1: Find the maximum output for each path.

$$O_{1,max} = \min\{M_i(p^{t_{j,i}+1} + \alpha_i p^{t_{j,i}+k+1} q) | i \in P_1\} \text{ and}$$
$$O_{2,max} = \min\{M_i(p^{t_{j,i}+1} + \alpha_i p^{t_{j,i}+k+1} q) | i \in P_2\}, \quad (10)$$

where $t_{j,i}$ is the number of follow-up machines after machine e_i on the general manufacturing path $P_j^{(G)}$. The indicator α_i equals 1 for the ith machine not on the reworking path; otherwise α_i equals 0.

Step 2: Find the maximum output of the common machine e_c.

$$O_{c,max} = M_c(p^{t_c+1} + \alpha_c p^{t_c+k+1} q), \quad (11)$$

where M_c is the maximal capacity of e_c, while t_c is the number of follow-up machines after e_c on the general manufacturing path $P_j^{(G)}$. The indicator α_c equals 1 for e_c not on the reworking path; otherwise α_c equals 0.

If $d \leqslant O_{c,max}$, go to Step 3; otherwise, stop (i.e., $d > O_{c,max}$ indicates that the manufacturing network can not satisfy such a demand level).

Step 3: Find the demand assignment (d_1, d_2) satisfying $d_1 + d_2 = d$ under constraints $d_1 \leqslant O_{1,max}$, $d_2 \leqslant O_{2,max}$.

Step 4: For each demand pair (d_1, d_2), do the following steps.

4.1. Determine the amount of input materials for each production line by

$$I_1 = d_1/(p^{n_1} + p^{n_1+k} q) \text{ and}$$
$$I_2 = d_2/(p^{n_2+1} + p^{n_2+1+k} q), \quad (12)$$

Note that, since the common machine e_c is installed in P_1, the number of machines in P_2 is less than P_1 and $n_1 = n_2 + 1$.

4.2. Determine the input flows for each machine e_i. For $e_i \in P_j$, the input flows are derived as follows:

$$f_{j,i}^{(G)} = I_j p^{s_{j,i}-1} \text{ for } i \text{ such that } e_i \in P_j^{(G)} \text{ and}$$
$$f_{j,i}^{(R|r,r-k)} = I_j p^{s_{j,i}+k-1} q \text{ for } i \text{ such that } e_i \in P_j^{(R|r,r-k)}, \quad (13)$$

where r is the sequence of the machine whose defective output can be reworked; $s_{j,i}$ is the sequence of e_i on the general manufacturing path $P_j^{(G)}$.

4.3. Transform input flows from general manufacturing paths and reworking paths into machines' loading vector $L = (l_1, l_2, \ldots, l_c, \ldots, l_{n_1+n_2})$ via

$$l_i = \sum_{j=1}^{2} \left(f_{j,i}^{(G)} + f_{j,i}^{(R|r,r-k)} \right), \quad (14)$$

where l_c is the loading of the common machine e_c.

Note: the minimal capacity that e_c should provide is determined by sum of loadings in both production lines, in which $l_c = f_{1,c}^{(G)} + f_{1,c}^{(R|r,r-k)} + f_{2,c}^{(G)} + f_{2,c}^{(R|r,r-k)}$.

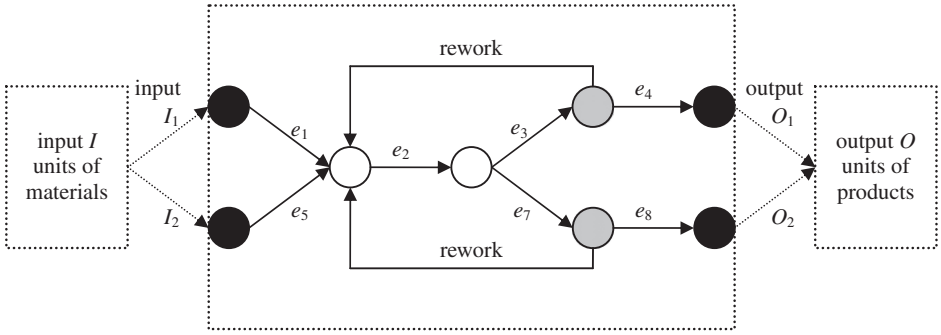


Fig. 4. Two distinct production lines for Scenario 2.

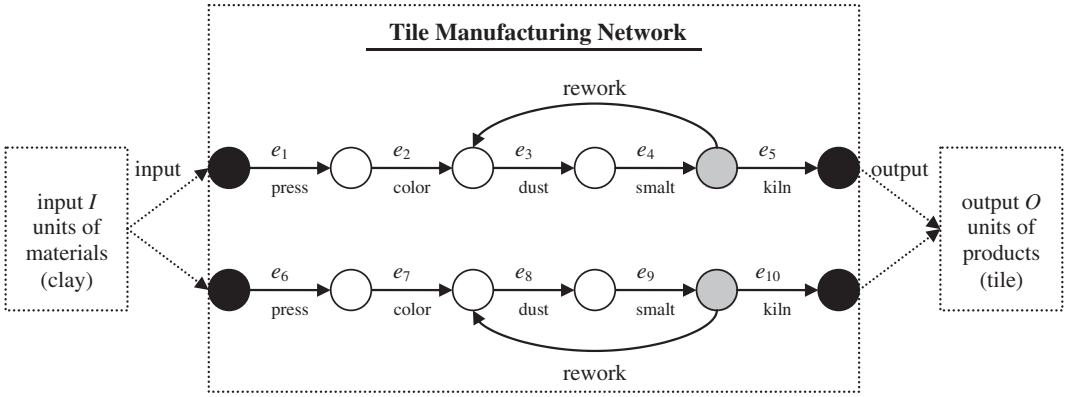


Fig. 5. Tile manufacturing network in Scenario 1.

4.4. For each machine, find the smallest possible capacity x_{ic} such that $x_{ic} \geqslant l_i > x_{i(c-1)}$. Then $Y = (y_1, y_2, \ldots, y_{n_1+n_2})$ is a minimal capacity vector for d, where $y_i = x_{ic}$ for all i.

Step 5: Those Y obtained in Step 4 are the minimal capacity vectors for d.

6. Numerical examples

A tile manufacturing system is utilized to demonstrate the proposed algorithms. The tile manufacturing system is a high-volume production of only a certain type of products. We model the tile manufacturing system as a manufacturing network as shown in Figs. 5 and 6 for different scenarios, respectively. The tiles (output products) are made starting from clay (input raw materials) and the regular process of tile manufacturing can be characterized as three steps, say pressing (e_1 and e_6), glazing, and burning (e_5 and e_{10}). For the glazing step, different machines put the colors (e_2 and e_7), dust (e_3 and e_7), and smalts (e_4 and e_9) on the WIP and then entering the burning kiln (Andres, Albarracin, Tormo, Vicens, & Garcia-Sabater, 2005). In the tile manufacturing network, defective WIP (smalts are put unevenly) output from the smalts machines e_4 and e_9 (the 4th sequenced machine in both production lines, $r = 4$) are reworked starting from the previous one ($k = 1$) machine (i.e., starting from e_3 and e_7, respectively). Thus, the tile manufacturing network can be described as a flow-shop manufacturing network with five machines for each production line. The success rate of each machine $p = 0.95$ and the machine data is given in Table 1. The capacity distribution in Table 1 is in the form of probability mass function, indicating that each machine can provide a certain capacity level with a corresponding probability. For instance, the probability of machine e_1 can produce 200 tiles (or more) per unit time is $\Pr\{x_1 \geqslant 200\} = 0.01 + 0.02 + 0.95 = 0.98$. We consider the tile manufacturing network in both scenarios has to satisfy demand $d = 400$ (tiles per unit time) and every 50 tiles are packaged in a box. The system reliabilities R_{400} for both scenarios are derived in following subsections.

6.1. Scenario 1

Consider the tile manufacturing network in Scenario 1 as shown in Fig. 5. Two identical paralleled production lines, say $P_1 = \{e_1, e_2, e_3, e_4, e_5\}$ and $P_4 = \{e_6, e_7, e_8, e_9, e_{10}\}$, are producing the same product type through five machines. By the decomposition method, the tile manufacturing is decomposed into four paths, say $P_1^{(G)} = \{e_1, e_2, e_3, e_4, e_5\}$, $P_1^{(R|4,3)} = \{e_3, e_4, e_5\}$, $P_2^{(G)} = \{e_6, e_7, e_8, e_9, e_{10}\}$, and $P_2^{(R|4,3)} = \{e_8, e_9, e_{10}\}$. The production routine of two lines is given in Table 2.

Step 1: Find the maximum output for each path.

$$\begin{aligned} O_{1,max} &= \min\{400 \times (0.95^{4+1} + 0.95^{4+1+1} \times 0.05), 600 \\ &\quad \times (0.95^{3+1} + 0.95^{3+1+1} \times 0.05), 300 \times (0.95^{2+1} \\ &\quad + 0), 400 \times (0.95^{1+1} + 0), 300 \times (0.95^{0+1} + 0)\} \\ &= \min\{324.214, 511.917, 257.213, 361.0, 285.0\} \\ &= 257.213, \end{aligned}$$

$$\begin{aligned} O_{2,max} &= \min\{405.268, 511.917, 257.213, 361.0, 342.0\} \\ &= 257.213. \end{aligned}$$

Step 2: Find the demand assignment (d_1, d_2) satisfying $d_1 + d_2 = 400$, $d_1 \leqslant 257.213$ and $d_2 \leqslant 257.213$. Since every 50 tiles are packaged in a box, the feasible demand pairs are $D_1 = (150, 250)$, $D_2 = (200, 200)$, and $D_3 = (250, 150)$.

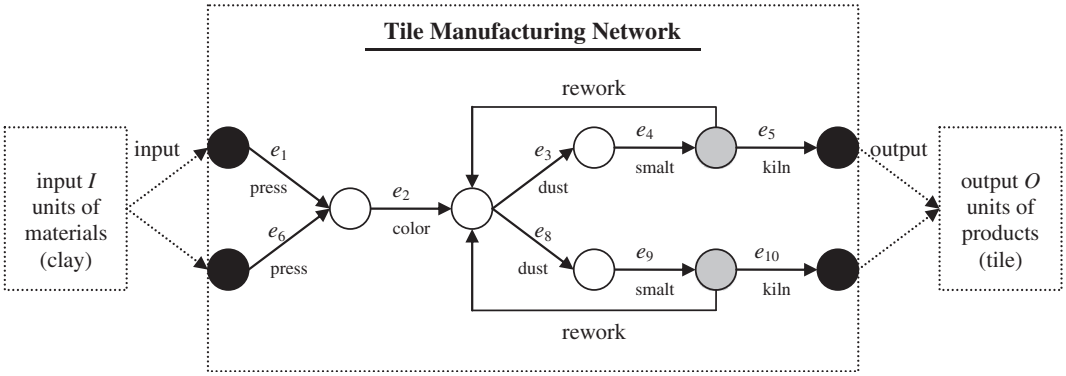


Fig. 6. Tile manufacturing network in Scenario 2.

Table 1
The machine data of example.

Machine	Capacity[a]	Probability	Machine	Capacity[a]	Probability
e_1	0	0.01	e_6	0	0.01
	100	0.01		100	0.01
	200	0.01		200	0.01
	300	0.02		300	0.02
	400	0.95		400	0.02
				500	0.93
e_2	0	0.005	e_7[b]	0	0.005
	150	0.01		150	0.005
	300	0.01		300	0.015
	450	0.015		450	0.015
	600	0.96		600	0.96
e_3	0	0.002	e_8	0	0.001
	50	0.003		50	0.002
	100	0.005		100	0.002
	150	0.01		150	0.005
	200	0.01		200	0.01
	250	0.015		250	0.015
	300	0.955		300	0.965
e_4	0	0.001	e_9	0	0.001
	50	0.001		50	0.001
	100	0.001		100	0.001
	150	0.002		150	0.002
	200	0.002		200	0.002
	250	0.003		250	0.003
	300	0.01		300	0.01
	350	0.01		350	0.01
	400	0.97		400	0.97
e_5	0	0.001	e_{10}	0	0.002
	60	0.002		60	0.003
	120	0.002		120	0.005
	180	0.005		180	0.005
	240	0.01		240	0.01
	300	0.98		300	0.01
				360	0.965

[a] Tiles that each machine can process per unit time.
[b] e_7 is not existing in Scenario 2.

Step 3: For each demand pair (d_1, d_2), do the following steps.
For $D_1 = (150, 250)$.

3.1a. The input materials of each production line is

$$I_1 = 150/(0.95^5 + 0.95^{5+1} \times 0.05) = 185.063 \text{ and}$$
$$I_2 = 250/(0.95^5 + 0.95^{5+1} \times 0.05) = 308.438.$$

Note that $n_1 = n_2 = 5$.

3.2a. For demand pair $D_1 = (150, 250)$, the input flow of each machine is shown in Table 3.

3.3a. Transform input flows from both general manufacturing path and reworking path into machines' loading vector $L_1 = (185.063, 175.810, 174.953, 166.205, 157.895, 308.438, 293.016, 291.588, 277.008, 263.158)$.

3.4a. For each machine, find the smallest possible capacity x_{ic} such that $x_{ic} \geqslant l_i > x_{i(c-1)}$. For demand pair $D_1 = (150, 250)$, the minimal capacity vector $Y_1 = (200, 300, 200, 200, 180, 400, 300, 300, 300, 300)$.
For $D_2 = (200, 200)$.

3.1b The input materials of each production line is

$$I_1 = 200/(0.95^5 + 0.95^{5+1} \times 0.05) = 246.750 \text{ and}$$
$$I_2 = 200/(0.95^5 + 0.95^{5+1} \times 0.05) = 246.750.$$

$\vdots$

Step 4: Three minimal capacity vectors for $d = 400$ are obtained from Step 3, say $Y_1 = (200, 300, 200, 200, 180, 400, 300, 300, 300, 300)$, $Y_2 = (300, 300, 250, 250, 240, 300, 300, 250, 250, 240)$, and $Y_3 = (400, 300, 300, 300, 300, 200, 300, 200, 200, 180)$.

In Scenario 1, three minimal capacity vectors for $d = 400$ are generated and thus we obtain the system reliability $R_{400} = 0.883199$ by the RSDP algorithm.

6.2. Scenario 2

Consider the tile manufacturing network in Scenario 2 as shown in Fig. 6. Two distinct production lines are producing two types of product, in which $P_1 = \{e_1, e_2, e_3, e_4, e_5\}$ is producing tiles with color and $P_2 = \{e_6, e_8, e_9, e_{10}\}$ is producing tiles without color. Note that the color machine e_7 is no more in P_2. However, the production manager decides to produce products with color by both production lines. Thus, the color machine e_2 in P_1 is set as a common machine (i.e., $e_c = e_2$). In the case, two production lines are $P_1 = \{e_1, e_2, e_3, e_4, e_5\}$ and $P_2 = \{e_6, e_2, e_8, e_9, e_{10}\}$ by using the same color machine e_2. By the decomposition method, the tile manufacturing is decomposed into four paths, say $P_1^{(G)} = \{e_1, e_2, e_3, e_4, e_5\}, P_1^{(R|4,3)} = \{e_3, e_4, e_5\}, P_2^{(G)} = \{e_6, e_2, e_8, e_9, e_{10}\}$, and $P_2^{(R|4,3)} = \{e_8, e_9, e_{10}\}$. The production routine of two lines is given in Table 4.

Step 1: Find the maximum output for each path.

$$O_{1,\max} = \min\{324.214, 511.917, 257.213, 361.0, 285.0\} = 257.213,$$
$$O_{2,\max} = \min\{405.268, 511.917, 257.213, 361.0, 342.0\} = 257.213.$$

Step 2: Find the maximum output of the common machine e_2.

Table 2
Production routine of the tile manufacturing network in Scenario 1.

line (j)	Machine # (e_i)	Sequence ($s_{j,i}$)	Follow-up machines	# Of follow-up machine ($t_{j,i}$)	Indicator (α_i)[a]
1	e_1	1	e_2, e_3, e_4, e_5	4	1
	e_2	2	e_3, e_4, e_5	3	1
	e_3	3	e_4, e_5	2	0
	e_4	4	e_5	1	0
	e_5	5	–	0	0
2	e_6	1	e_7, e_8, e_9, e_{10}	4	1
	e_7	2	e_8, e_9, e_{10}	3	1
	e_8	3	e_9, e_{10}	2	0
	e_9	4	e_{10}	1	0
	e_{10}	5	–	0	0

[a] $\alpha_i = 1$ For the ith machine not on the reworking path; otherwise $\alpha_i = 0$.

Table 3
The results of Step 3 for $(d_1,d_2) = (150, 250)$ in Scenario 1.

Input flow	i				
	1	2	3	4	5
$f_{1,i}^{(G)}$	185.063	175.810	167.019	158.668	150.735
$f_{1,i}^{(R\|4,3)}$	0	0	7.933	7.537	7.160
$l_i = f_{1,i}^{(G)} + f_{1,i}^{(R\|4,3)}$	185.063	175.810	174.953	166.205	157.895
y_i	200	300	200	200	180
	6	7	8	9	10
$f_{2,i}^{(G)}$	308.438	293.016	278.365	264.447	251.225
$f_{2,i}^{(R\|4,3)}$	0	0	13.222	12.561	11.933
$l_i = f_{2,i}^{(G)} + f_{2,i}^{(R\|4,3)}$	308.438	293.016	291.588	277.008	263.158
y_i	400	300	300	300	300

$$O_{c,\max} = 600(0.95^{3+1} + 0.95^{3+1+1}0.05) = 511.917,$$
$$d = 400 \leqslant O_{c,\max} = 511.917, \text{ go to Step 3.}$$

Step 3: Find the demand assignment (d_1,d_2) satisfying $d_1 + d_2 = 400$, $d_1 \leqslant 257.213$ and $d_2 \leqslant 257.213$. Since every 50 tiles are packaged in a box, the feasible demand pairs are $D_1 = (150, 250)$, $D_2 = (200, 200)$, and $D_3 = (250, 150)$.

Step 4: For each demand pair (d_1, d_2), do the following steps.
For $D_1 = (150, 250)$.

4.1a. The input materials of each production line is

$$I_1 = 150/(0.95^5 + 0.95^{5+1} \times 0.05) = 185.063 \text{ and}$$
$$I_2 = 250/(0.95^{4+1} + 0.95^{4+1+1} \times 0.05) = 308.438.$$

Note that $n_1 = 5$ and $n_2 = 4$.

4.2a. For demand pair $D_1 = (150, 250)$, the input flow of each machine is shown in Table 5.

4.3a. Transform input flows from both general manufacturing path and reworking path into the machines' loading vector $L_1 = (185.063, 175.810 + 293.016, 174.953, 166.205, 157.895, 308.438, 291.588, 277.008, 263.158)$.
Note: the minimal capacity that e_2 should provide is determined by sum of loadings in both production lines, which is $l_2 = 175.810 + 293.016 = 468.826$.

4.4a. For each machine, find the smallest possible capacity x_{ic} such that $x_{ic} \geqslant l_i > x_{i(c-1)}$. For demand pair $D_1 = (150, 250)$, the minimal capacity vector $Y_1 = (200, 600, 200, 200, 180, 400, 300, 300, 300)$.
For $D_2 = (200, 200)$.

4.1b. The input materials of each production line is

$$I_1 = 200/(0.95^5 + 0.95^{5+1} \times 0.05) = 246.750 \text{ and}$$
$$I_2 = 200/(0.95^5 + 0.95^{4+1+1} \times 0.05) = 246.750.$$

$\vdots$

Step 5: Three minimal capacity vectors for $d = 400$ are obtained from Step 4, say $Y_1 = (200, 600, 200, 200, 180, 400, 300, 300, 300)$, $Y_2 = (300, 600, 250, 250, 240, 300, 250, 250, 240)$, and $Y_3 = (400, 600, 300, 300, 300, 200, 200, 200, 180)$, respectively.

In Scenario 2, three minimal capacity vectors for $d = 400$ are generated and thus we obtain the system reliability $R_{400} = 0.869476$ by the RSDP algorithm.

6.3. Discussion on the results

Firstly, we concentrate on performance of the manufacturing network in both scenarios. Comparing the results in these two scenarios; it is significant that Scenario 1 performs better system reliability than Scenario 2 by 0.013723. In both scenarios, the

Table 4
Production routine of the tile manufacturing network in Scenario 2.

Line (j)	Machine # (e_i)	Sequence ($s_{j,i}$)	Follow-up machines	# Of follow-up machine ($t_{j,i}$)	Indicator (α_i)[a]
1	e_1	1	e_2, e_3, e_4, e_5	4	1
	e_2[b]	2	e_3, e_4, e_5	3	1
	e_3	3	e_4, e_5	2	0
	e_4	4	e_5	1	0
	e_5	5	–	0	0
2	e_6	1	e_2, e_8, e_9, e_{10}	4	1
	e_2[b]	2	e_8, e_9, e_{10}	3	1
	e_8	3	e_9, e_{10}	2	0
	e_9	4	e_{10}	1	0
	e_{10}	5	–	0	0

[a] $\alpha_i = 1$ For the ith machine not on the reworking path; otherwise $\alpha_i = 0$.
[b] e_2 is a common machine.

Table 5
The results of Step 3 for $(d_1,d_2) = (150, 250)$ in Scenario 2.

Input flow	i				
	1	2	3	4	5
$f_{1,i}^{(G)}$	185.063	175.810	167.019	158.668	150.735
$f_{1,i}^{(R\|4,3)}$	0	0	7.933	7.537	7.160
$l_i = f_{1,i}^{(G)} + f_{1,i}^{(R\|4,3)}$	185.063	175.810	174.953	166.205	157.895
y_i	200	–[a]	200	200	180
	6	2	8	9	10
$f_{2,i}^{(G)}$	308.438	293.016	278.365	264.447	251.225
$f_{2,i}^{(R\|4,3)}$	0	0	13.222	12.561	11.933
$l_i = f_{2,i}^{(G)} + f_{2,i}^{(R\|4,3)}$	308.438	293.016	291.588	277.008	263.158
y_i	400	–[a]	300	300	300

[a] The minimal capacity of e_2 should be determined by sum of loadings in both production lines.

probability distributions of machines' capacity are the same. However, in Scenario 2, e_7 no longer exists in P_2. Therefore, to satisfy demand level $d = 400$ in both scenarios, e_2 who operates the same function as e_7 has to provide higher capacity in Scenario 2 than in Scenario 1. Thus, the system reliability is lower in Scenario 2.

The proposed algorithms can be easily executed for a specified demand pair. That is, the production manager could evaluate the satisfaction probability for each demand pair in terms of its corresponding minimal capacity vector. This probability is denoted as R_{d_1,d_2} for each demand pair, where $d_1 + d_2 = d$. Take Scenario 1 in the Example for instance, we obtain three minimal capacity vectors for $d = 400$. For demand pair $D_1 = (150, 250)$, the corresponding minimal capacity vector is $Y_1 = (200, 300, 200, 200, 180, 400, 300, 300, 300, 300)$ and the satisfaction probability $R_{150,250} = \Pr\{X|X \geqslant Y_1\} = 0.820466$. Table 6 shows R_{d_1,d_2} for different demand pairs. The results indicate that $D_2 = (200, 200)$ with $R_{200,200} = 0.838654$ would be a better way to produce products since it has higher satisfaction probability. Thus, the production manger may decide to produce 200 units of products per units of time for each production line in Scenario 1.

7. Generic model

We further discuss an extension case of more than two production lines. Thus, a more generic model is extended to deal with manufacturing systems with multiple production lines, which is the case in most of the applications. We extend Scenario 2 into a generic model. Note that, Scenario 1 can be a special case (without common machine) of Scenario 2. Hence, we modify Algorithm II for multiple production lines as follows.

7.1. Algorithm III: For multiple production lines

Suppose there are u production lines, say $P_1, P_2, \ldots, P_u$, in a manufacturing system. Each production line P_j consists of n_j machines, where $j = 1, 2, \ldots, u$. When producing the same product type, some machines, say $e_{c1}, e_{c2}, \ldots, e_{cm}$, are utilized as common machines. For each production line, defective WIP output from the rth sequenced machine is reworked starting from the previous k machine (i.e., from the $(r-k)$ th sequenced machine). The minimal capacity vectors for d can be derived as following steps.

Step 1: Find the maximum output for each path P_j, where $j = 1, 2, \ldots, u$.

$$O_{j,\max} = \min\{M_i(p^{t_{j,i}+1} + \alpha_i p^{t_{j,i}+k+1} q) | i \in P_j\}, \quad (15)$$

Table 6
Demand pair and corresponding R_{d_1,d_2}.

Demand pair	Minimal capacity vector	Probability
$D_1 = (150, 250)$	(200, 300, 200, 200, 180, 400, 300, 300, 300, 300)	$R_{150,250} = 0.820466$
$D_2 = (200, 200)$	(300, 300, 250, 250, 240, 300, 300, 250, 250, 240)	$R_{200,200} = 0.838654$
$D_3 = (250, 150)$	(400, 300, 300, 300, 300, 200, 300, 200, 200, 180)	$R_{250,150} = 0.820312$

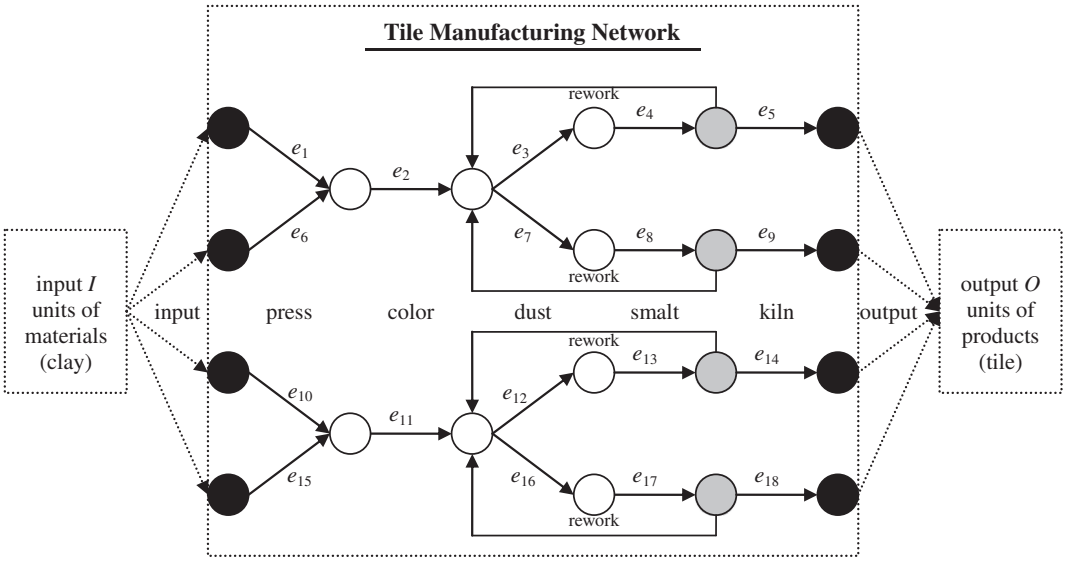


Fig. 7. Tile manufacturing network with multiple production lines.

Table 7
The machine data in multiple production lines case.

Machine	Capacity[a]	Probability	Machine	Capacity[a]	Probability
e_1/e_{10}	0	0.01	e_6/e_{15}	0	0.01
	100	0.01		100	0.01
	200	0.01		200	0.01
	300	0.02		300	0.02
	400	0.95		400	0.02
				500	0.93
e_2/e_{11}	0	0.005			
	150	0.01			
	300	0.01			
	450	0.015			
	600	0.96			
e_3/e_{12}	0	0.002	e_7/e_{16}	0	0.001
	50	0.003		50	0.002
	100	0.005		100	0.002
	150	0.01		150	0.005
	200	0.01		200	0.01
	250	0.015		250	0.015
	300	0.955		300	0.965
e_4/e_{13}	0	0.001	e_8/e_{17}	0	0.001
	50	0.001		50	0.001
	100	0.001		100	0.001
	150	0.002		150	0.002
	200	0.002		200	0.002
	250	0.003		250	0.003
	300	0.01		300	0.01
	350	0.01		350	0.01
	400	0.97		400	0.97
e_5/e_{14}	0	0.001	e_9/e_{18}	0	0.002
	60	0.002		60	0.003
	120	0.002		120	0.005
	180	0.005		180	0.005
	240	0.01		240	0.01
	300	0.98		300	0.01
				360	0.965

[a] Tiles that each machine can process per unit time.

where $t_{j,i}$ is the number of follow-up machines after machine e_i on the general manufacturing path $P_j^{(G)}$. The indicator α_i equals 1 for the ith machine not on the reworking path; otherwise α_i equals 0.

Step 2: Find the maximum output of the vth common machine e_{cv}, where $v = 1, 2, \ldots, m$.

$$O_{cv,max} = M_{cv}(p^{t_{cv}+1} + \alpha_{cv}p^{t_{cv}+k+1}q), \tag{16}$$

where M_{cv} is the maximal capacity of machine e_{cv}, while t_{cv} is the number of follow-up machines after e_{cv} on the general manufacturing path $P_j^{(G)}$. The indicator α_{cv} equals 1 for e_{cv} not on the reworking path; otherwise α_{cv} equals 0.
If $\sum_{j:e_{cv}\in P_j} d_j \leqslant O_{cv,max}$, go to Step 3; otherwise, stop (i.e., $\sum_{j:e_{cv}\in P_j} d_j > O_{cv,max}$ indicates that the manufacturing network can not satisfy such a demand level).

Step 3: Find the demand assignment $(d_1, d_2, \ldots, d_u)$ satisfying $\sum_{j=1}^{u} d_j = d$ under constraints $d_j \leqslant O_{j,max}$.

Step 4: For each demand assignment $(d_1, d_2, \ldots, d_u)$, do the following steps.

4.1. Determine the amount of input materials for each production line by

$$I_j = d_j/(p^{n_j} + p^{n_j+k}q). \tag{17}$$

4.2. Determine the input flows for each machine e_i. For $e_i \in P_j$, the input flows are derived as follows:

$$f_{j,i}^{(G)} = I_j p^{s_{j,i}-1} \text{ for } i \text{ such that} e_i \in P_j^{(G)} \text{ and}$$
$$f_{j,i}^{(R|r,r-k)} = I_j p^{s_{j,i}+k-1} q \text{ for} i \text{ such that } e_i \in P_j^{(R|r,r-k)}, \tag{18}$$

where r is the sequence of the machine whose defective output can be reworked; $s_{j,i}$ is the sequence of e_i on the general manufacturing path $P_j^{(G)}$.

4.3. Transform input flows from general manufacturing paths and reworking paths into machines' loading vector $L = (l_1, l_2, \ldots, l_{n_1+n_2+\ldots+n_u})$ via

$$l_i = \sum_{j=1}^{u} \left(f_{j,i}^{(G)} + f_{j,i}^{(R|r,r-k)} \right). \tag{19}$$

4.4. For each machine, find the smallest possible capacity x_{ic} such that $x_{ic} \geqslant l_i > x_{i(c-1)}$. Then $Y = (y_1, y_2, \ldots, y_{n_1+n_2+\ldots+n_j})$ is a minimal capacity vector for d, where $y_i = x_{ic}$ for all i.

Step 5: Those Y obtained in Step 4 are the minimal capacity vectors for d.

7.2. Example of multiple production lines

Consider the tile manufacturing network with multiple production lines as shown in Fig. 7. Four production lines, say $P_1 = \{e_1, e_2, e_3, e_4, e_5\}$, $P_2 = \{e_6, e_2, e_7, e_8, e_9\}$, $P_3 = \{e_{10}, e_{11}, e_{12}, e_{13}, e_{14}\}$, and $P_4 = \{e_{15}, e_{11}, e_{16}, e_{17}, e_{18}\}$, are producing the same product type with two common machines e_2 and e_{11}. The success rate of each machine $p = 0.95$ and the machine data is given in Table 7. In this example, 125 minimal capacity vectors for $d = 400$ are generated by Algorithm III and thus we obtain the system reliability $R_{400} = 0.997439$ by the RSDP algorithm. The results indicate that $D_{57} = (50, 150, 50, 150)$ with $R_{50,150,50,150} = 0.854712$ would be a better way to produce products since it has higher satisfaction probability.

8. Conclusion

To assess the performance of a manufacturing system, this paper develops a stochastic-flow network model considering multiple production lines. We model the manufacturing system as a manufacturing network by the proposed graphical-based transformation and decomposition. We evaluate the possibility that a manufacturing network can provide sufficient capacities to satisfy customers' order. Thus, the probability that a manufacturing network could produce d units of products with reworking actions, named system reliability, is evaluated. We take two different scenarios into account: Scenario 1 that two identical production lines in parallel producing the same product type; Scenario 2 that two distinct production lines producing the same product type. In each Scenario, we find the minimal capacity vectors that machines should provide to satisfy demand d. In terms of such vectors, the system reliability is derived by applying the RSDP algorithm. Furthermore, we provide a more generic model to deal with manufacturing systems with multiple production lines. The proposed models can be easily extended to the case that each machine possesses a distinct yield rate. That is, we can replace the same yield rate p by a distinct yield rate p_i of machine e_i. For instance, input I units of raw materials, the products produced by general processing path through machines $e_1, e_2, \ldots, e_n$ is $I\prod_{t=1}^{n} p_t$.

We also address the decision making issue based on the minimal capacity vectors derived. The production manager can determine a better way to assign the amount of output that each production line should produce to fulfill demand. According to the system reliability, the production manager could conduct a sensitivity analysis to investigate the most important machine in a manufacturing network to improve the system to be more reliable.

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